

4	(a)	(i)	Resultant force acts perpendicular to the velocity of the object. / Centripetal force is constant in magnitude	B1
			Marker's comment Generally well done. Some students were vague when referring to "the centre" without specifying whether it is the centre of the circular motion or the centre of the object.	
		(ii)	The velocity of the object is always changing as the <u>direction</u> of the motion is always <u>changing</u> . Thus the object experiences a rate of change of velocity. Acceleration is in the direction of the resultant force / perpendicular to the velocity / towards the centre of the circular path.	B1 B1
	(b)	(i)	The gravitational force provides the centripetal force to keep the satellite in circular motion. Thus, gravitational force = mv^2/r $GMm/r^2 = mv^2/r$ $r = \frac{GM}{v^2} = \frac{(6.67 \times 10^{-11})(6.0 \times 10^{24})}{3700^2}$ $r = 2.92 \times 10^7 \text{ m}$	B1 C1 A1
			Marker's comment For (i), students should explain that the direction of velocity changes which leads to a change in velocity which means that there is acceleration. For (ii), the majority could obtain the correct answer.	
		(ii)	$v = r\omega = r \frac{2\pi}{T}$ $T = \frac{2\pi r}{v} = 4.96 \times 10^5 \text{ s} = 13.8 \text{ h}$ Since the time is not the same period as the rotation of the Earth (24 h), the satellite is not a geostationary satellite	M1 A1
			Marker's comment Students who obtained full credit presented a systematic presentation of steps - find the period of the given satellite - comparing this period with 24 hours (the period of a geostationary satellite) - making a conclusion ECF would have been given to those who used their value of r in (i), even though is was incorrect. However, credit could not be given to answers that stated the period of a geostationary satellite to be 1 year.	

		(iii)	$GPE = -GMm/r = -2.74 \times 10^7 \text{ J}$ $KE = \frac{1}{2}mv^2 = 1.37 \times 10^7 \text{ J}$ Total energy = GPE + KE = $-1.37 \times 10^7 \text{ J}$	C1 C1 A1
		Marker's comment Many students mistakenly ignored the negative sign in the GPE expression. There were also students who merely used the expression $-GMm/2r$ to obtain the correct answer. They received partial credit. Students are reminded that this expression has to be derived before use.		
	(c)	(i)	Any mass ejected in the same direction as the satellite. (Idea of rocket motor pushing against direction of motion of satellite.	B1
		Marker's comment This question was poorly done. The force in question has to slow down the satellite and thus must act against its instantaneous velocity.		
		(ii)	Kinetic energy of the satellite in a circular orbit is inversely proportional to the radius of the circular orbit. ($KE = GMm/2r$) With a smaller orbital radius, kinetic energy of the satellite increases.	C1 A1
		Marker's comment There were many wrong concepts found in students' answers here. Many believed that total energy remained the same despite r decreasing and so mistakenly quoted conservation of mechanical energy. Others believed that the centripetal force remained constant in different orbits. Correct responses relied on the expression $KE = GMm/2r$ or $v^2 = GM/r$ to determine the dependence of KE with r . If faced with similar questions in future, students are advised to look for a mathematical expression to guide their thinking.		

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